

Patient-level inputs

SLD trajectory

$y_{i,j}$ at weeks $t_{i,j}$

Baseline covariates

$\mathbf{x}_i$ (clinical, biomarker, ctDNA)

Event history

progression, death, dropout

Mechanistic submodel

Two-component state-space SLD

$D_i(t)$, $G_i(t)$ (latent log-states)

rates r_i^{dec} , r_i^{gro} | $\mathbf{x}_i$

Time-varying covariates

$\mathbf{W}_i(t) = (\log B_i(t), \log r_i^{\text{dec}}, \log r_i^{\text{gro}})$

Multistate submodel

Illness-death + dropout

$\lambda_{jk}(t | \mathbf{W}_i(t), \mathbf{x}_i)$

with GP baseline hazards

Posterior endpoints

PFS

composite &
cause-specific

OS

illness-death
routing

Response & other

ORR, hazard ratios,
landmark survival

joint likelihood